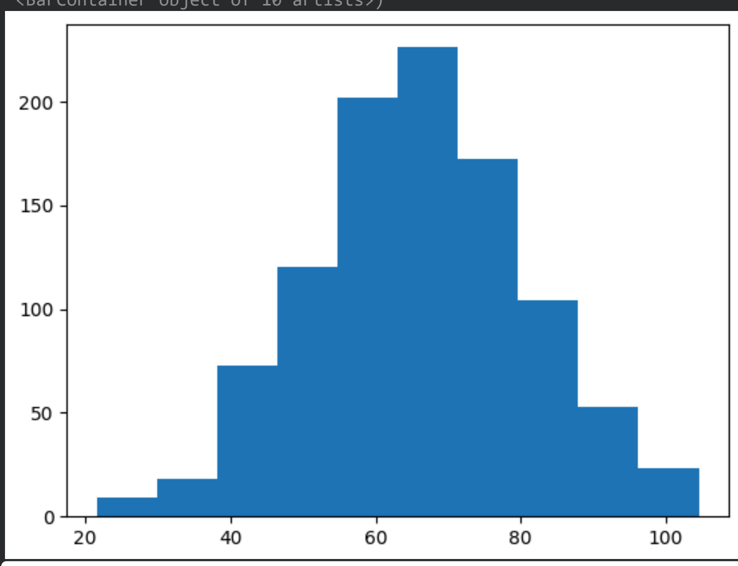
```
import numpy as np
import pandas as pd

# Set a random seed for reproducible results
np.random.seed(42)

# Generate 1,000 rows of mock customer data
data = {
    "Customer_ID": range(1, 1001),
    "Age": np.random.randint(18, 70, size=1000),  # Uniform distribution
    "Annual_Income_K": np.random.normal(
        65, 15, size=1000
    ).round(1),  # Normal distribution
    "Spending_Score": np.random.randint(1, 100, size=1000),
}

# Create the DataFrame
df = pd.DataFrame(data)

# Display the first 5 rows
print(df.head())
```

```
   Customer_ID  Age  Annual_Income_K  Spending_Score
0            1   56             40.9              58
1            2   69             68.1               4
2            3   46             53.7               4
3            4   32             43.7              20
4            5   60             55.3              10
```

```
plt.hist(df['Annual_Income_K'])
```

```
(array([  9.,  18.,  73., 120., 202., 226., 172., 104.,  53.,  23.]),
 array([ 21.6 ,  29.89,  38.18,  46.47,  54.76,  63.05,  71.34,  79.63,
         87.92,  96.21, 104.5 ]),
 <BarContainer object of 10 artists>)
```



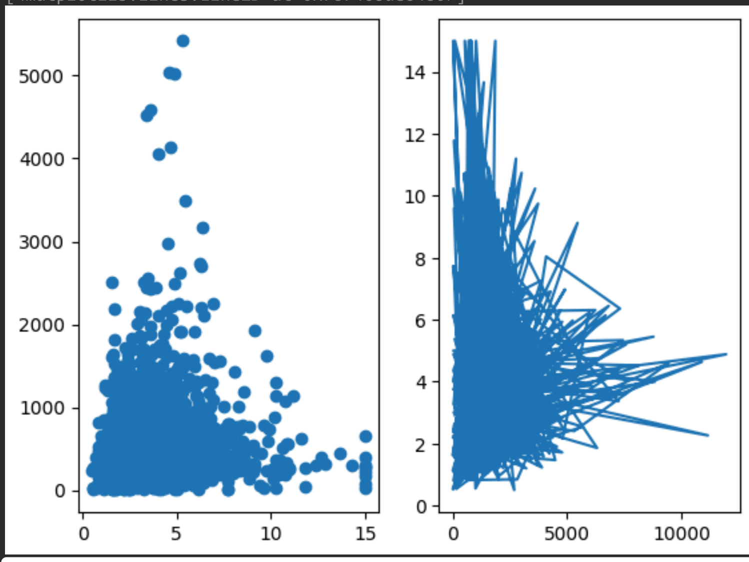

```
import pandas as pd
df = pd.read_csv('/content/sample_data/california_housing_test.csv')
```

```
df.head()
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_valu
0	-122.05	37.37	27.0	3885.0	661.0	1537.0	606.0	6.6085	344700
1	-118.30	34.26	43.0	1510.0	310.0	809.0	277.0	3.5990	176500
2	-117.81	33.78	27.0	3589.0	507.0	1484.0	495.0	5.7934	270500
3	-118.36	33.82	28.0	67.0	15.0	49.0	11.0	6.1359	330000
4	-119.67	36.33	19.0	1241.0	244.0	850.0	237.0	2.9375	81700

```
plt.subplot(1,2,1)
plt.scatter(df['median_income'], df['total_bedrooms'])
plt.subplot(1,2,2)
plt.plot(df['population'], df['median_income'])
```

```
[<matplotlib.lines.Line2D at 0x787468de6480>]
```



```
#
x = [-30, 1,2,3,4,5,6,7,8,9,10,11,1,2,13,14, 100, 105, 99]
plt.boxplot(x)
```

```
{'whiskers': [<matplotlib.lines.Line2D at 0x7874682a41a0>,
  <matplotlib.lines.Line2D at 0x7874682a4470>],
 'caps': [<matplotlib.lines.Line2D at 0x7874682a4620>,
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 'means': []}
```

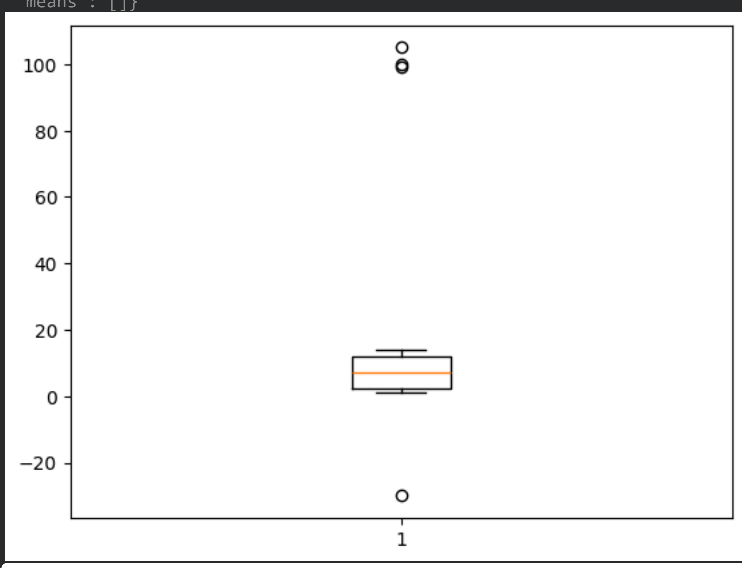

```
# Heatmap, Boxplot detailed, Stats mean, median & mode
# How to detect outliers
# How to handle the outliers.
```

```
import pandas as pd
df = pd.read_csv('/content/train.csv')
df
```

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	F
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
1455	1456	60	RL	62.0	7917	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	
1456	1457	20	RL	85.0	13175	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	M
1457	1458	70	RL	66.0	9042	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	G
1458	1459	20	RL	68.0	9717	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	
1459	1460	20	RL	75.0	9937	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	

1460 rows × 81 columns

```
import matplotlib.pyplot as plt
# df['SalePrice']
plt.boxplot(df['SalePrice'])
```

```
{'whiskers': [<matplotlib.lines.Line2D at 0x7d0ac11796d0>,
  <matplotlib.lines.Line2D at 0x7d0ac11788f0>],
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 'fliers': [<matplotlib.lines.Line2D at 0x7d0ac117bb60>],
 'means': []}
```
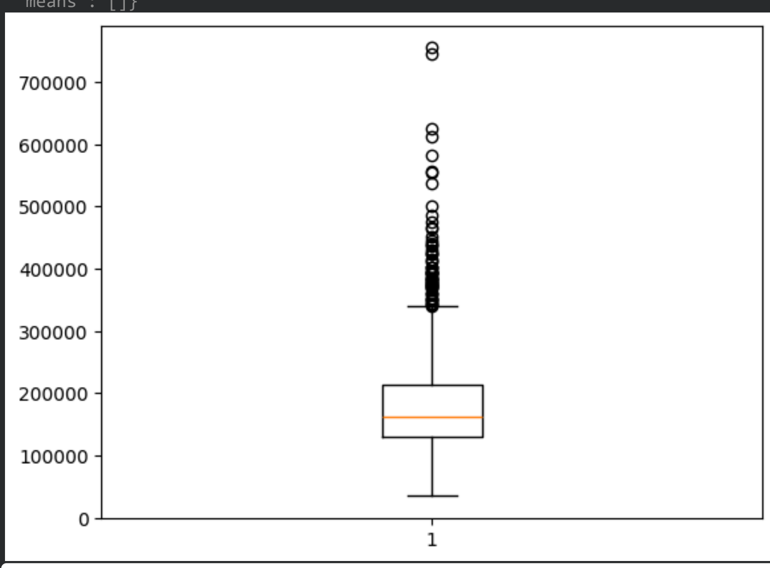




```
Q1 = df['SalePrice'].quantile(0.25)
Q3 = df['SalePrice'].quantile(0.75)
IQR = Q3 - Q1 #Inter Quartile range
upper = Q3 + 1.5 * IQR
lower = Q1 - 1.5 * IQR
# (df['SalePrice'] < upper )
filtered_data = df[(df['SalePrice'] > lower) & (df['SalePrice'] < upper )]
```

```
num_df.columns
```

```
Index(['Id', 'MSSubClass', 'LotFrontage', 'LotArea', 'OverallQual',
       'OverallCond', 'YearBuilt', 'YearRemodAdd', 'MasVnrArea', 'BsmtFinSF1',
       'BsmtFinSF2', 'BsmtUnfSF', 'TotalBsmtSF', '1stFlrSF', '2ndFlrSF',
       'LowQualFinSF', 'GrLivArea', 'BsmtFullBath', 'BsmtHalfBath', 'FullBath',
       'HalfBath', 'BedroomAbvGr', 'KitchenAbvGr', 'TotRmsAbvGrd',
       'Fireplaces', 'GarageYrBlt', 'GarageCars', 'GarageArea', 'WoodDeckSF',
       'OpenPorchSF', 'EnclosedPorch', '3SsnPorch', 'ScreenPorch', 'PoolArea',
```

```
       'MiscVal', 'MoSold', 'YrSold', 'SalePrice'],
      dtype='object')
```

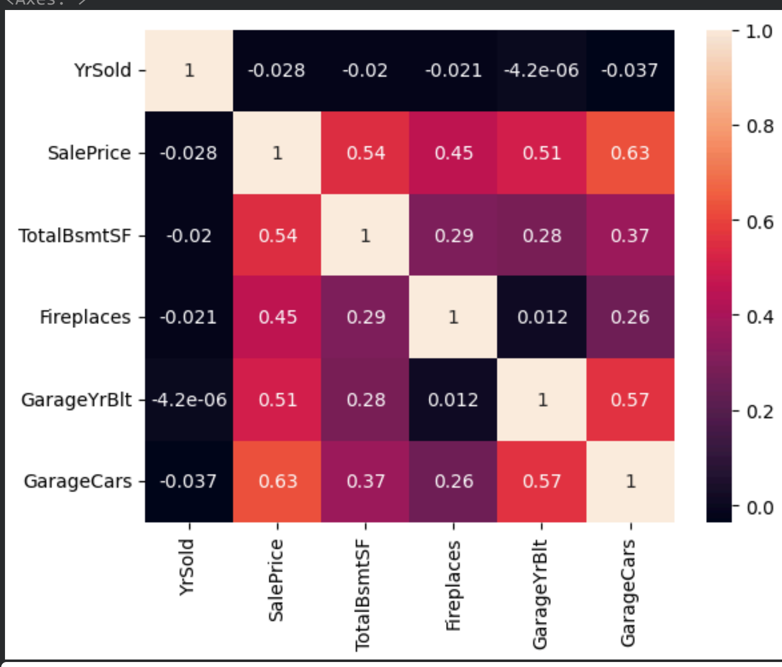
```
# EDA -> Exploratory Data Analysis
# Univariate analysis
# Bivariate Analysis
# Multivariate analysis
```

```
num_df = filtered_data.select_dtypes(exclude='object')
list_col= ['YrSold', 'SalePrice','TotalBsmtSF','Fireplaces', 'GarageYrBlt', 'GarageCars']
num_df[list_col].corr()
```

	YrSold	SalePrice	TotalBsmtSF	Fireplaces	GarageYrBlt	GarageCars
YrSold	1.000000	-0.028245	-0.020029	-0.021330	-0.000004	-0.036707
SalePrice	-0.028245	1.000000	0.543508	0.453010	0.507894	0.628013
TotalBsmtSF	-0.020029	0.543508	1.000000	0.293132	0.279432	0.373523
Fireplaces	-0.021330	0.453010	0.293132	1.000000	0.012048	0.258011
GarageYrBlt	-0.000004	0.507894	0.279432	0.012048	1.000000	0.566266
GarageCars	-0.036707	0.628013	0.373523	0.258011	0.566266	1.000000

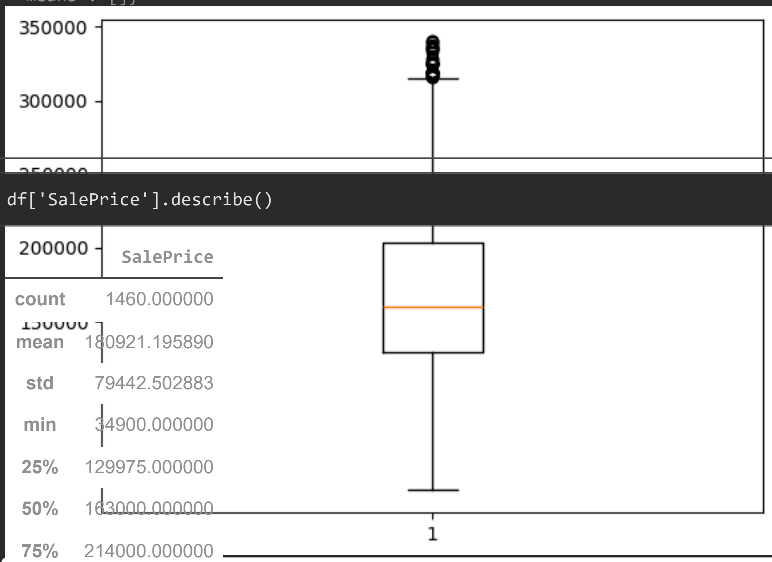
```
import seaborn as sns
sns.heatmap(num_df[list_col].corr(), annot=True)
```

<Axes: >




```
import matplotlib.pyplot as plt
# df['SalePrice']
plt.boxplot(filtered_data['SalePrice'])
```

```
{'whiskers': [<matplotlib.lines.Line2D at 0x7d0ac0f854f0>,
  <matplotlib.lines.Line2D at 0x7d0ac0f85910>],
 'caps': [<matplotlib.lines.Line2D at 0x7d0ac0f85c40>,
  <matplotlib.lines.Line2D at 0x7d0ac0f85f40>],
 'boxes': [<matplotlib.lines.Line2D at 0x7d0ac117be90>],
 'medians': [<matplotlib.lines.Line2D at 0x7d0ac0f86360>],
 'fliers': [<matplotlib.lines.Line2D at 0x7d0ac0f866f0>],
 'means': []}
```




```
df['SalePrice'].describe()
```

	SalePrice
count	1460.000000
mean	180921.195890
std	79442.502883
min	34900.000000
25%	129975.000000
50%	163000.000000
75%	214000.000000
max	755000.000000

dtype: float64

```
x = [1,2,3]
y = [10, 20, 30]
plt.subplot(1,2,1)
plt.plot(x, y)
plt.subplot(1,2,2)
plt.bar(x, y)
```

```
<BarContainer object of 3 artists>
```